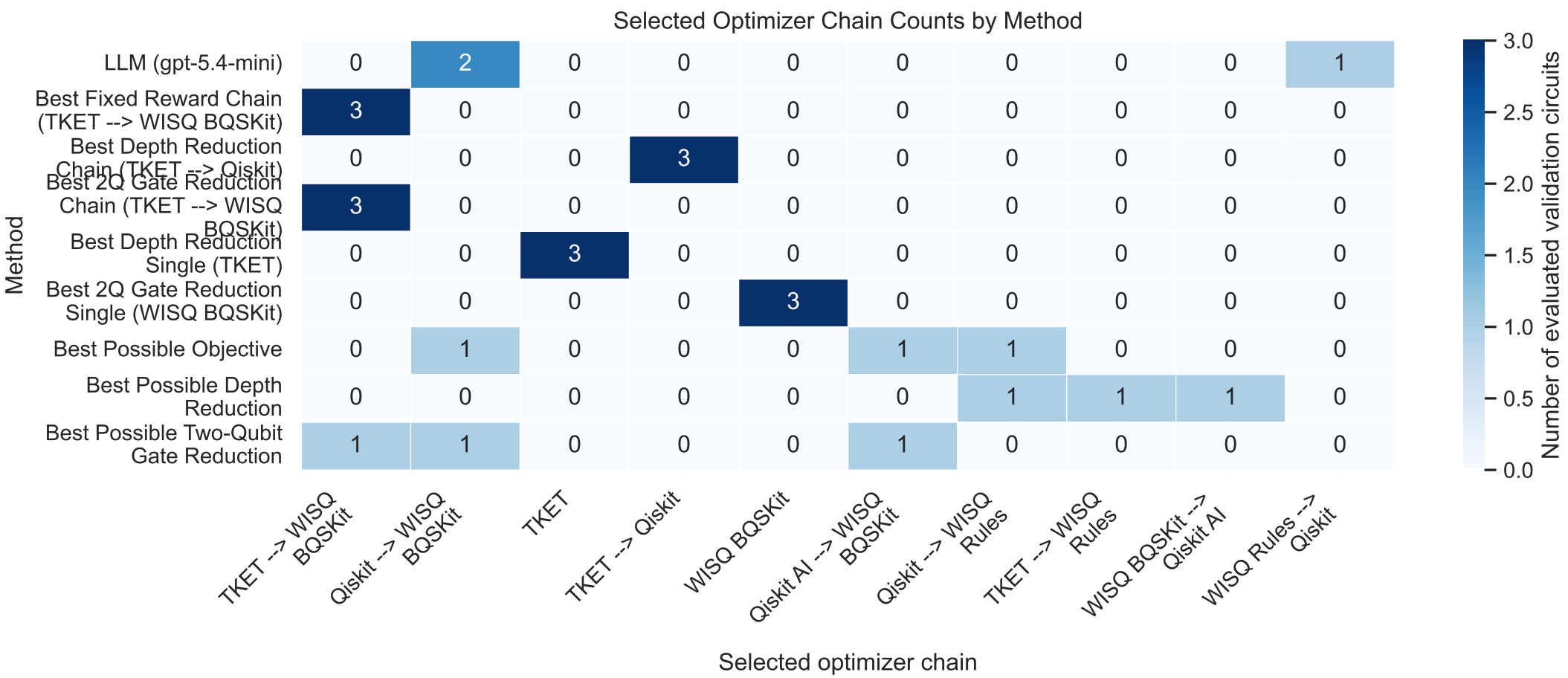
Selected Optimizer Chain Counts by Method
Method
LLM (gpt-5.4-mini)
Best Fixed Reward Chain (TKET --> WISQ BQSKit)
Best Depth Reduction Chain (TKET --> Qiskit)
Best 2Q Gate Reduction Chain (TKET --> WISQ BQSKit)
Best Depth Reduction Single (TKET)
Best 2Q Gate Reduction Single (WISQ BQSKit)
Best Possible Objective
Best Possible Depth Reduction
Best Possible Two-Qubit Gate Reduction
0 2 0 0 0 0 0 0 0 1
3 0 0 0 0 0 0 0 0 0
0 0 0 3 0 0 0 0 0 0
3 0 0 0 0 0 0 0 0 0
0 0 3 0 0 0 0 0 0 0
0 0 0 0 3 0 0 0 0 0
0 1 0 0 0 1 1 0 0 0
0 0 0 0 0 0 1 1 1 0
1 1 0 0 0 1 0 0 0 0
TKET --> WISQ BQSKit
Qiskit --> WISQ BQSKit
TKET
TKET --> Qiskit
WISQ BQSKit
Qiskit AI --> WISQ BQSKit
Qiskit --> WISQ Rules
TKET --> WISQ Rules
WISQ BQSKit --> Qiskit AI
WISQ Rules --> Qiskit
Selected optimizer chain
Number of evaluated validation circuits
3.0
2.5
2.0
1.5
1.0
0.5
0.0